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Display apparatus

Abstract

A display apparatus is disclosed that includes a display panel having front and rear surfaces; a heat dissipation plate having front and rear surfaces and the rear surface of the heat dissipation plate is on the rear surface of the display panel; a source printed circuit board on the rear surface of the heat dissipation plate, and connected to the display panel; a control printed circuit board on the rear surface of the heat dissipation plate and comprising a retaining protrusion that protrudes towards the rear surface of the display panel; and a holder member between the heat dissipation plate and the control printed circuit board and holding the control printed circuit board in the heat dissipation plate. The control printed circuit board is stably held and thus expected to have effects on reducing shaking during logistics transportation and simplifying a manufacturing process.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Republic of Korea Patent Application No. 10-2022-0175944, filed on Dec. 15, 2022, which is incorporated by reference in its entirety.

BACKGROUND

Field of the Invention

(2) The disclosure relates to a display apparatus.

Description of the Related Art

(3) A display apparatus may, for example, be a liquid crystal display (LCD) apparatus, an organic light emitting display (OLED) apparatus, an inorganic light emitting display apparatus, etc.

(4) The display apparatus such as the OLED apparatus including a self-emissive device has been widespread to various fields such as a dashboard, a windshield, a mirror display, and indoor and outdoor billboards, etc. of a vehicle as well as traditional electronic devices such as a television (TV). In this case, a use environment for each display apparatus is required to be optimized.

SUMMARY

- (5) An aspect of the disclosure is to provide a display apparatus of which a control printed circuit board is stably held to minimize or at least reduce shaking during logistics transportation and simplify a manufacturing process.
- (6) It will be appreciated that problems to be solved according to an embodiment of the disclosure are not limited to those mentioned above, and other problems not mentioned will become apparent to those skilled in the art from the following description.
- (7) In one embodiment, a display apparatus comprises: a display panel including a front surface and a rear surface that is opposite the front surface; a heat dissipation plate including a front surface and a rear surface, the front surface of the heat dissipation plate on the rear surface of the display panel; a source printed circuit board on the rear surface of the heat dissipation plate, the source printed circuit board connected to the display panel via a flexible circuit board; a control printed circuit board on the rear surface of the heat dissipation plate and connected to the source printed circuit board via a flat cable, the control printed circuit board comprising a retaining protrusion that protrudes towards the rear surface of the display panel; and a holder member between the heat dissipation plate and the control printed circuit board, the holder member including at least one through hole through which the retaining protrusion is inserted.
- (8) In one embodiment, a display apparatus comprises: a display panel including a front surface and a rear surface that is opposite the front surface; a heat dissipation plate including a front surface and a rear surface and the front surface of the heat dissipation plate is on the rear surface of the display panel, the heat dissipation plate configured to dissipate heat from the display panel; a holder frame on the heat dissipation plate, the holder frame including a plurality of through holes; and a control printed circuit board on the holder frame, the control printed circuit board including a plurality of retaining protrusions that are each inserted into a corresponding through hole from the plurality of through holes and in contact with at least a portion of the corresponding through hole such that the control printed circuit board is connected to the holder frame.
- (9) According to an embodiment of the disclosure, a control printed circuit board is stably held to minimize or at least reduce shaking during logistics transportation, thereby preventing or at least reducing damage and noise.
- (10) Further, a tape for holding is not used, thereby reducing manufacturing costs.
- (11) Further, a control printed circuit board is provided to a customer as it has already been held in the heat dissipation plate without a repetitive screw fastening process, thereby simplifying an assembly process.
- (12) According to an embodiment of the disclosure, a display apparatus is prevented from defects such as shaking of a printed circuit board, thereby enhancing a yield and optimizing a process based on less consumption of production energy.
- (13) According to an embodiment of the disclosure, a space for dissipating heat is formed in a holder frame of a display apparatus to improve heat dissipation, thereby reducing operation failure due to heat and providing a low-power display apparatus.
- (14) The foregoing problems to be solved, the means for solving the problems, and the effects do not specify the essential features of claims, and thus the scope of the claims is not limited by the matters disclosed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram showing the front of a display apparatus according to an embodiment of the disclosure.
- (2) FIG. 2 is a diagram showing a cross section of a light emitting device according to an embodiment of the disclosure.

- (3) FIG. 3 is a perspective view showing the rear of a display apparatus according to an embodiment of the disclosure.
- (4) FIG. 4 is a plan view showing the rear of a display apparatus according to an embodiment of the disclosure.
- (5) FIG. 5A is a cross-sectional view showing a portion in which a second protrusion is inserted in an insertion hole in a holder member according to an embodiment of the disclosure.
- (6) FIG. 5B is a perspective view showing a portion in which a second protrusion is inserted in an insertion hole in a holder member according to an embodiment of the disclosure.
- (7) FIG. 6A is a cross-sectional view showing that a recessed portion is inserted in a retaining hole in a holder member according to an embodiment of the disclosure.
- (8) FIG. 6B is a perspective view showing that a recessed portion is inserted in a retaining hole in a holder member according to an embodiment of the disclosure.
- (9) FIG. 7 is a perspective view of a holder frame according to an embodiment of the disclosure.
- (10) FIG. 8 shows a cross-sectional structure taken along line A-A of FIG. 3 according to an embodiment of the disclosure.
- (11) FIG. 9 is a plan view of a holder member according to an embodiment of the disclosure.
- (12) FIG. 10 is a plan view of a holder member according to another embodiment of the disclosure.
- (13) FIG. 11 is a plan view of a holder member according to still another embodiment of the disclosure.

DETAILED DESCRIPTION

- (14) Advantages and features of the disclosure and methods of accomplishing the same may be understood more readily by reference to the detailed description of embodiments that will be made hereinafter with reference to the accompanying drawings. The disclosure may, however, be embodied in many different forms and should not be construed as being limited to the exemplary embodiments set forth herein; rather, these exemplary embodiments are provided so that the disclosure will be thorough and complete and will fully convey the concept of the invention to those skilled in the art, and the present invention will only be defined by the appended claims.
- (15) The shapes, sizes, ratios, angles, numbers and the like illustrated in the drawings to describe embodiments of the disclosure are merely exemplary, and thus, the disclosure is not limited thereto. Throughout the specification, the same reference numerals refer to the same components. In addition, detailed descriptions of well-known technologies may be omitted in the disclosure to avoid obscuring the subject matter of the disclosure. When terms such as “comprises,” “has,” “includes,” or “is made up of” are used in this specification, it should be understood that unless “only” is specifically used, additional elements or steps can be included. Unless otherwise explicitly stated, when a component is expressed in the singular form, it is intended to encompass the plural form as well.
- (16) In interpreting the components, it is construed to include a margin of error even in the absence of explicit description.
- (17) When describing the positional relationship, for example, when the relationship between two parts is described as “on”, “on top of”, “underneath”, “beside”, etc., unless “directly” or “immediately” is used, one or more other parts may be located between the two parts.
- (18) When a device or layer is referred to as being “on” another device or layer, it includes cases where one device or layer is directly located on the other device or layer or still other device or layer is interposed between the two devices or layers.
- (19) Although the terms “first”, “second”, and the like are used to describe various components, these components are not limited by these terms. These terms are merely used for distinguishing one component from the other components. Therefore, the first component mentioned hereinafter may be the second component in the technical sense of the disclosure.
- (20) Throughout the specification, the same reference numerals refer to the same components.
- (21) The sizes and thicknesses of each component shown in the drawings are presented for the

convenience of description and are not intended to limit the disclosure.

(22) The features of various embodiments of the disclosure can be partially or entirely combined or assembled in various technical manners of interlocking and interoperation obvious to those skilled in the art, and each embodiment can be implemented independently or in combination with related embodiments.

(23) Hereinafter, detailed descriptions are made of the embodiments of the disclosure with reference to the accompanying drawings.

(24) In the disclosure, the term “display apparatus” is used in a narrow sense to refer to display apparatuses, such as a liquid crystal module (LCM), an organic light-emitting diode (OLED) module, and a quantum dot module, each including a display panel and a panel driving unit to operate the display panel. In addition, the term may also be used to refer to set electronic devices, set devices, or apparatuses that include LCMs, OLED modules, QD modules, and the like, such as equipment display apparatuses including complete or final products such as laptop computers, televisions, computer monitors, automotive displays or equipment displays provided in other forms for vehicles, as well as mobile electronic devices such as smartphones or electronic pads.

(25) Accordingly, in the disclosure, the display apparatus may include not only the display apparatuses in the narrow sense themselves, such as LCMs, OLED modules, QD modules, but also set devices as application products or final consumer devices including LCMs, OLED modules, QD modules, and the like.

(26) Additionally, in some cases, an LCM, an OLED module, or a QD module, composed of a display panel and a panel driving unit may be referred to as a “display apparatus” in a narrow sense, while an electronic device as a complete product including an LCM, an OLED module, or a QD module may be referred to as “set device”. For example, the narrow-sense display apparatus may include a display panel of liquid crystal (LCD), organic light-emitting diode (OLED), or quantum dot (QD) and a source printed circuit board (PCB) as a control unit for driving the display panel, while a set device may further include a set PCB, serving as a set control unit that is electrically connected to the source PCB and controls the entire set device.

(27) The display panel used in the embodiments may include all types of display panels such as liquid crystal display panels, organic light-emitting diode (OLED) display panels, quantum dot (QD) display panels, and electroluminescent display panels, and is not limited to a specific display panel capable of bending a bezel with a flexible substrate for the OLED display panel and a backplate support structure thereunder. In addition, the display panel used in the embodiments of the disclosure is not limited to the shape or size of the display panel.

(28) For example, when the display panel is an organic light-emitting diode (OLED) display panel, it may include a plurality of gate lines and data lines and pixels formed at the intersection of the gate lines and data lines. In addition, it may be configured to include an array including thin-film transistors as components for selectively applying a voltage to each pixel, an organic light-emitting diode (OLED) layer on the array, an encapsulation substrate or encapsulation layer arranged on the array to cover the organic light-emitting diode layer, etc. The encapsulation layer may protect the thin film transistors and the organic light-emitting device layer from external impacts and prevent moisture or oxygen from penetrating the organic light-emitting device layer. In addition, the layers formed on the array may, for example, include an inorganic light-emitting layer, such as a nano-sized material layer or quantum dots.

(29) In the disclosure, FIG. 1 illustrates an exemplary organic light emitting diode (OLED) display panel that may be integrated within display apparatuses.

(30) FIG. 1 is a diagram illustrating the front of a display apparatus **100** according to an embodiment of the disclosure.

(31) With reference to FIG. 1, the display apparatus **100** may be applied to the center fascia of a vehicle. Although FIG. 1 shows a rectangular display panel **150**, the shape of the display apparatus **100** is not necessarily limited thereto and can be produced in various shapes such as squares,

polygons, or curves.

(32) Referring to FIG. 1, the display apparatus **100** is equipped with integrated display and touch functions, and the cover member **101** is attached to the front. A display panel **150** for displaying may be arranged on the back surface of the cover member **101**. The display apparatus **100** may be inserted and coupled to the dashboard of a vehicle, for example. The display panel **150** may be made on a substrate of a rigid material, but in the case of organic light-emitting devices, it may also be made on a flexible substrate. A display panel **150** made on a flexible substrate may be deformed concavely or convexly depending on the shape of the dashboard, allowing for flexible design. The cover member **101** may be a cover glass.

(33) FIG. 2 shows the cross-sectional structure of the light emitting device provided in the display panel of FIG. 1 according to one embodiment. The substrate **111** may support various components of the display panel **150**. The substrate **111** may be formed of a transparent dielectric material such as glass, plastic, and the like. In the case of being made of plastic, the substrate **111** may be a plastic film or a plastic substrate. For example, the substrate **111** may take the form of a film including one of the polyimide-based polymers, polyesters-based polymers, silicone-based polymers, acrylic-based polymers, polyolefin-based polymers, and their copolymers. Among these materials, polyimide is mainly used as a plastic substrate because it is suitable for high-temperature processes and is a material that can be coated.

(34) A buffer layer may be positioned on the substrate **111**. The buffer layer is a functional layer that protects the thin film transistor (TFT) from impurities such as alkali ions that may leak from the bottom of the substrate **111**. The buffer layer may be formed of silicon oxide (SiOx), silicon nitride (SiNx), or multiple layers thereof.

(35) Thin film transistor **130** may be disposed on the buffer layer. The thin film transistor **130** may be formed by sequentially arranging a gate electrode **132**, a gate insulating layer **112**, a semiconductor layer **134**, an interlayer insulating film **114**, a source electrode **136** and a drain electrode **138** on the buffer layer. There may be one or more thin film transistors **130** arranged for a plurality of sub-pixels provided in the active area.

(36) Although illustrated as the bottom-gate type in FIG. 2, the thin film transistor **130** is not limited thereto and may also be provided as the top-gate type, in which the order of the semiconductor layer **134** and the gate electrode **132** are reversed. The disclosure is not limited thereto.

(37) The semiconductor layer **134** may be arranged at a specific portion on the substrate **111** or on the buffer layer. The semiconductor layer **134** may be made of polycrystalline silicon (p-Si), and in this case, a region of the semiconductor layer **134** may be doped with impurities to form the electrode layer. The semiconductor layer **134** may also be made of amorphous silicon (a-Si) and various organic semiconductor materials such as pentacene. Furthermore, the semiconductor layer **134** may also be made of an oxide material. The gate insulating layer **112** may be formed of inorganic dielectric materials such as silicon oxide (SiOx) or silicon nitride (SiNx) and organic dielectric materials. The gate electrode **132** may be formed of various conductive materials, such as magnesium (Mg), aluminum (Al), nickel (Ni), chromium (Cr), molybdenum (Mo), tungsten (W), gold (Au), or alloys thereof.

(38) The first interlayer insulating film **114** may be formed of inorganic dielectric materials such as silicon oxide (SiOx) or silicon nitride (SiNx) and, organic dielectric materials. The first interlayer insulating film **114** may be selectively removed to form contact holes exposing the source and drain regions.

(39) The source and drain electrodes **136** and **138** are formed as a single layer or multilayer of electrode material on the first interlayer insulating film **114**.

(40) An inorganic protective film **116** and a planarization layer **118** may be positioned on the thin film transistor **130** to cover the source and drain electrodes **136** and **138**. The inorganic protective film **116** and the planarization layer **118** protect the thin film transistor **130** and flatten its upper

surface.

(41) The inorganic protective film **116** may be formed of inorganic dielectric films such as silicon nitride (SiNx) and silicon oxide (SiOx), while the planarization layer **118** may be made of organic dielectric films such as Benzocyclobutene (BCB) or acrylic (Acryl). The inorganic protective film **116** and the planarization layer **118** may each be formed as a single layer, dual-layer, or multilayer structure, and in some cases, one of the two layers may be omitted.

(42) A light-emitting component OLED connected to the thin film transistor (TFT) **130** may be formed by sequentially arranging a first electrode **122**, an organic light-emitting layer **124**, and a second electrode **126**. That is, the light-emitting component OLED may be composed of the first electrode **122** connected to the drain electrode **138** through a connection hole **148** formed in the planarization layer **118** and the inorganic protective film **116**, the organic light emitting layer **124** positioned on the first electrode **122**, and the second electrode **126** positioned on the organic light-emitting layer **124**.

(43) When the display panel **150** is of a top emission type where the emission occurs upward through the second electrode **126**, the first electrode **122** may include an opaque conductive material with high reflectivity. In this case, examples of reflective conductive material may include silver (Ag), aluminum (Al), gold (Au), molybdenum (Mo), tungsten (W), chromium (Cr), or alloys thereof.

(44) A bank **128** is formed in the area excluding the light-emitting area, opening up the light-emitting area. Accordingly, the bank **128** has a bank hole exposing the first electrode **122** corresponding to the light-emitting area. The bank **128** may be made of inorganic dielectric materials such as silicon nitride (SiNx), silicon oxide (SiOx), or organic dielectric materials such as BCB, acrylic-based resins, or imide-based resins.

(45) The organic light-emitting layer **124** is positioned on the first electrode **122** exposed by the bank **128**. The organic light-emitting layer **124** may include a hole injection layer, a hole transport layer, an emissive layer, an electron transport layer, and an electron injection layer. In addition, the organic light-emitting layer **124** may be composed of a single emissive layer structure that emits a single light within a single stack, or a multi-stack structure including multiple stacks, each of which includes a single emissive layer of the same color. In such cases, adjacent sub-pixels may be arranged to emit different colors of light to display various colors. For example, sub-pixels with emissive layers of red, green, and blue may be arranged in a row or spaced apart from each other and, as well, in a triangle shape or pentile structure with some sub-pixels of predetermined colors aligned parallel and others aligned diagonally to each other.

(46) In some cases, sub-pixels of white color may also be added to the arrangement. In addition, the organic light-emitting layer **124** may be configured by stacking a plurality of stacks including emissive layers emitting different colors of light to express white. In the case of expressing white with a stacked structure, separate color filters may be additionally added to each sub-pixel.

(47) The second electrode **126** is positioned on the organic light-emitting layer **124**. When the display panel **150** adopts a top emission structure, the second electrode **126** may be formed of a transparent conductive material such as indium tin oxide (ITO) or indium zinc oxide (IZO) or a semi-transparent metal or metal alloy such as Mg and Ag to emit light generated in the organic light-emitting layer **124** upward through the second electrode **126**.

(48) The second electrode **126** may be arranged to reach a capping layer. The capping layer may protect the OLED and assist in the extraction of light emitted through the second electrode **126** by using a material with a high refractive index.

(49) An encapsulation layer **140** may be disposed on the light-emitting component OLED. The encapsulation layer **140** prevents or at least reduces the infiltration of oxygen and moisture from the outside, preventing oxidation of the emissive and electrode materials. Exposure of OLED to moisture or oxygen may cause pixel shrinkage or the formation of dark spots, reducing the emitting area. The encapsulation layer **140** is formed by alternately stacking inorganic layers **142** and **146**

made of glass, metal, aluminum oxide (AlOx), or silicon (Si) materials and an organic layer **144** that serves as a buffer relieving the stress between layers due to the bending of the display panel (**150** in FIG. **1**) and enhances the flattening performance. The organic layer **144** may be made of organic dielectric materials such as acrylic resin, epoxy resin, polyimide, polyethylene, or silicon oxycarbide (SiOC). Here, the first and second inorganic layers **142** and **146** serve to block the penetration of moisture or oxygen, while the organic layer **144** flattens the surface of the first inorganic layer **142**. The encapsulation layer **140** is composed of several thin film layers to increase the length and complexity of the path that moisture and oxygen must travel compared to a single layer, with the purpose of making it difficult for moisture and oxygen to penetrate to the light-emitting component OLED.

(50) A protective layer may be formed further between the light-emitting component OLED and the encapsulation layer **140** to protect the encapsulation layer **140** from being peeled off or affecting the uniformity during the manufacturing process of the encapsulation layer **140**.

(51) With reference to FIG. **2**, a touch buffer layer **154** may be disposed on the encapsulation layer **140**. The touch buffer layer **154** may minimize or at least reduce the impact of light entering the display panel **150** from external light sources on the semiconductor layer **134** or the organic light-emitting layer **124**.

(52) With reference to FIG. **2**, a touch sensor layer **155** may be arranged on the touch buffer layer **154**. The touch sensor layer **155** may be structured with the first touch electrode **155a** and the second touch electrode **155c** intersecting each other, allowing one electrode to receive an applied voltage signal and the other to sense the voltage signal. The first touch electrode **155a** and the second touch electrode **155c** may be patterned into polygonal or circular shapes on a touch insulation film **155b** to be arranged at a distance from each other.

(53) The cover member **101** may be disposed on the touch sensor layer **155** (not shown in FIG. **2**).

(54) FIGS. **3** and **4** shows the rear of the display apparatus **100** according to an embodiment of the disclosure.

(55) FIG. **3** is a perspective view showing the rear of a display apparatus according to an embodiment of the disclosure.

(56) FIG. **4** is a plan view showing the rear of a display apparatus according to an embodiment of the disclosure.

(57) Referring to FIGS. **1**, **3** and **4**, the display apparatus **100** may include a cover member **101**, a display panel **150**, a heat dissipation plate **160**, a source printed circuit board **170**, a control printed circuit board **180**, and a holder member **200**.

(58) The cover member **101** may include an active area AA (see FIG. **1**) and an inactive area NA (see FIG. **1**). The source printed circuit board **170** may be attached to the back surface of the display panel **150**. The heat dissipation plate **160** may be disposed between a rear surface of the display panel **150** that is opposite a front surface of the display panel **150** and the source printed circuit board **170**. The heat dissipation plate **160** may be provided to dissipate the heat generated in the display panel to the rear of the display panel.

(59) The source printed circuit board **170** and the display panel **150** may be connected through a flexible circuit board **175** such that the source printed circuit board **170** can operate the display panel **150**. The flexible circuit board **175** may be a flexible circuit, a chip on film (COF), or the like.

(60) The control printed circuit board **180** and the source printed circuit board **170** may be connected through a flat cable **185** such that the control printed circuit board **180** can operate the source printed circuit board **170**. Here, the source printed circuit board **170** may be provided with a pair of first connectors **171**, and the control printed circuit board **180** may be provided with a pair of second connectors **181**. The flat cable **185** may be connected to the first connector **171** and the second connector **181** to connect together the control printed circuit board **180** and the source printed circuit board **170**.

(61) The heat dissipation plate **160** may be disposed on the back surface of the display panel **150** to dissipate the heat generated by the display panel **150**. In an embodiment of the disclosure, the heat dissipation plate **160** is an aluminum plate, but is not limited thereto.

(62) The holder member **200** may be disposed between the heat dissipation plate **160** and the control printed circuit board **180**. The holder member **200** may be disposed in connection with the heat dissipation plate **160** and the control printed circuit board **180**, so that the control printed circuit board **180** can be coupled to the heat dissipation plate **160**.

(63) The control printed circuit board **180** may include a retaining protrusion **240** in one embodiment.

(64) The holder member **200** may include a holder frame **210** and a through hole **230**. In one embodiment, the retaining protrusion **240** passes through the through hole **230**.

(65) The holder member **200** may be disposed on the rear of the heat dissipation plate **160**.

(66) Referring to FIG. 4, the holder frame **210** may be shaped corresponding to the control printed circuit board **180**. According to an embodiment of the disclosure, the control printed circuit board **180** is shaped like a quadrangular plate, and thus the holder frame **210** is shaped like a corresponding quadrangular plate. That is, the control printed circuit board **180** and the holder frame **210** have a same shape.

(67) The holder frame **210** may be implemented with a non-conductive and highly rigid material and disposed being adhered to the rear of the heat dissipation plate **160** by an adhesive member **190** (see FIG. 5a). For example, the holder frame **210** may be made of a plastic material, a ceramic material, or the like material, and may also be made of other materials.

(68) FIG. 7 is a perspective view of a holder frame according to an embodiment of the disclosure.

(69) Referring to FIG. 7, the holder frame **210** may be formed with an opening **220** and the through hole **230** thereon. The holder member **200** may include the holder frame **210** including the through hole **230**. In one embodiment, the holder frame **210** includes a plurality of through holes **230** that are each disposed at a corner of the holder frame **210**.

(70) The opening **220** may be formed by penetrating through an entire thickness of a center portion of the holder frame **210**. In this case, the opening **220** may be formed on the holder frame **210** in an area corresponding to (e.g., overlapping) the control printed circuit board **180**. While the display apparatus **100** is operating, the control printed circuit board **180** may generate heat. When the center portion of the holder frame **210** is blocked, heat does not dissipate and the control printed circuit board **180** heats up. This may deteriorate the performance of the display apparatus **100**.

(71) FIG. 8 shows a perspective view of a cross-sectional structure taken along line A-A of FIG. 3 showing the rear of the display apparatus according to one embodiment.

(72) Referring to FIG. 8, heat generated by the control printed circuit board **180** is indicated by the arrows. The heat is introduced into the inside of the holder frame **210** through the opening **220** formed in the center portion of the holder frame **210**. In this way, the holder frame **210** has a space to which heat is released (e.g., dissipated), and the heat dissipation plate **160** having high thermal conductivity is disposed so that heat generated in the control printed circuit board **180** can be easily dissipated to the outside by heat conduction in the heat dissipation plate **160**. The opening **220** forms an air gap between the control printed circuit board **180** and the heat dissipation plate **160** therein to prevent the heat generated in the control printed circuit board **180** from being transferred to the display panel **150**, thereby having an insulating effect. Accordingly, operation failure due to overheating is reduced.

(73) Referring to FIG. 7, the through hole **230** may be disposed through an entire thickness of the holder frame **210**. In this case, a plurality of through holes **230** may be disposed on the edges of the holder frame **210**. According to an embodiment of the disclosure, four through holes **230** are disposed to stably hold the edges of the control printed circuit board **180**, but not limited thereto.

(74) Referring to FIGS. 5A to 7 and 9 to 11, the through hole **230** according to an embodiment of the disclosure may include an insertion hole **231**, a holding portion **236**, and a retaining hole **233**.

(75) FIG. 5A is a cross-sectional view showing a portion in which a second protrusion is inserted in an insertion hole in a holder member **210** according to an embodiment of the disclosure.

(76) FIG. 5B is a perspective view showing a portion in which a second protrusion is inserted in an insertion hole in a holder member according to an embodiment of the disclosure.

(77) FIG. 6A is a cross-sectional view showing that a recessed portion is inserted in a retaining hole in a holder member according to an embodiment of the disclosure.

(78) FIG. 6B is a perspective view showing that a recessed portion is inserted in a retaining hole in a holder member according to an embodiment of the disclosure.

(79) FIGS. **9**, **10** and **11** are plan views of a holder member according to embodiments of the disclosure.

(80) Referring to FIG. 6B, the insertion hole **231** is disposed in the holder frame **210** and allows the retaining protrusion **240** for holding the control printed circuit board **180** to the holder frame **210** to be inserted into the insertion hole **231**. In addition, the retaining hole **233** is disposed in the holder frame **210** and extends from the insertion hole **231**, and holds the retaining protrusion **240**. In this case, the size of the retaining hole **233** may be smaller than that of the insertion hole **231** so that the retaining protrusion **240** is secured to the holder frame **210** within the retaining hole **233**.

(81) Further, a direction where the insertion hole **231** and the retaining hole **233** are arranged may correspond to a direction where the flat cable **185** connects the control printed circuit board **180** and the source printed circuit board **170**.

(82) Referring to FIG. 4, the flat cable **185** is disposed in the Y-axis direction while connecting the control printed circuit board **180** and the source printed circuit board **170**.

(83) Therefore, when the insertion hole **231** and the retaining hole **233** are arranged to extend in the Y-axis direction and the retaining protrusion **240** is inserted in the through hole **230** and moved downward into the retaining hole **233** and retained in the Z-axis direction (downward direction), the flat cable **185** stably connects the control printed circuit board **180** and the source printed circuit board **170** without being damaged.

(84) If the insertion hole **231** and the retaining hole **233** are arranged in the X-axis direction, the control printed circuit board **180** needs to move from side to side in the X-axis direction so as to be held, thereby causing the flat cable **185** to move from side to side (in the X-axis direction) and be twisted or damaged.

(85) Referring to FIGS. **9** to **11**, the holding portion **236** is disposed between the insertion hole **231** and the retaining hole **233** and formed to protrude inwards. The holding portion **236** may be formed to be smaller than a recessed portion **262**. The recessed portion **262** may be formed to be larger than the holding portion **236**. The holding portion **236** may interfere with the movement of the recessed portion **262** from the retaining hole **233** toward the insertion hole **231**.

(86) Referring to FIG. 5A to 6B, the retaining protrusion **240** may be disposed in the control printed circuit board **180** and detachably mounted to the through hole **230**.

(87) The retaining protrusion **240** may include a first protrusion **250** and a second protrusion **260**.

(88) The first protrusion **250** has a first end that is connected to a bottom portion of the control printed circuit board **180** and has a second end that protrudes in a downward direction away from the control printed circuit board **180**. In this case, the first protrusion **250** may include a coupling portion **252** at the first end that protrudes upwards. The coupling portion **252** may be fitted, welded, etc., to a hole formed in the control printed circuit board **180**. According to an embodiment of the disclosure, the first protrusion **250** and the coupling portion **252** may be made of a metal material including iron, steel, or the like, and may be a non-flexible material.

(89) The second protrusion **260** may be connected to the bottom of the first protrusion **250** and inserted in the through hole **230**.

(90) Referring to FIG. 6A, the second protrusion **260** according to an embodiment of the disclosure may include the recessed portion **262** and a bottom portion **264**. That is, the second protrusion **260** has a first end that is connected to the second end (e.g., the bottom) of the first protrusion **250**, a

second end (e.g., the bottom protrusion **264**) that is inserted in the through hole **230**, and the recessed portion **262** that is between the first end and the second end.

(91) The recessed portion **262** may be recessed inwards in the middle of the second protrusion **260**. Thus, a width of the recessed portion **262** is less than a width of the remaining portions of the recessed portion **262**. Further, the bottom portion **264** may be provided in the bottom of the second protrusion **260** and formed to be wider than the recessed portion **262**.

(92) Referring to FIG. **9**, the recessed portion **262** may be formed to have a width corresponding to the retaining hole **233**. Further, the bottom portion **264** is smaller than the insertion hole **231** but larger than the retaining hole **233**.

(93) According to an embodiment of the disclosure, the first and second protrusions **250** and **260** of the holder member **200** may be formed to have a circular cross-section. The cross-section of the first and second protrusions **250** and **260** according to embodiments of the disclosure may have various shapes such as a triangle shape, but is not limited thereto. A diameter $D_{sub.2}$ of the recessed portion **262** may be the same as or slightly smaller than a diameter $L_{sub.2}$ of the retaining hole **233**. The diameter $D_{sub.2}$ of the recessed portion **262** may correspond to the diameter $L_{sub.2}$ of the retaining hole **233**. This is to allow the recessed portion **262** to be inserted in the retaining hole **233**.

(94) The bottom portion **264** may be shaped corresponding to the shape of the insertion hole **231**, and the recessed portion **262** may be shaped corresponding to the retaining hole **233**.

(95) A diameter $D_{sub.1}$ of the bottom portion **264** may be smaller than a diameter $L_{sub.1}$ of the insertion hole **231**. This is to allow the bottom portion **264** to be smoothly inserted in the insertion hole **231**. Further, the diameter $D_{sub.1}$ of the bottom portion **264** may be larger than the diameter $L_{sub.2}$ of the retaining hole **233**. This is to make the bottom portion **264** interfere with the retaining hole **233** and prevent the control printed circuit board **180** from being separated from the holder frame **210** even though the control printed circuit board **180** is lifted upwards after the recessed portion **262** is inserted in the retaining hole **233**. This is to couple the control printed circuit board **180** to the holder frame **210**.

(96) Referring to FIG. **6A**, after the recessed portion **262** is inserted in the retaining hole **233**, the control printed circuit board **180** is not separated from the holder frame **210** even though the control printed circuit board **180** is lifted upwards because the bottom portion **264** has a larger diameter than the recessed portion **262** and thus interferes with the retaining hole **233**.

(97) Referring to FIG. **7**, the through holes **230** are disposed at four corners of the holder frame **210**. Referring to FIG. **4**, the retaining protrusions **240** are disposed at four corners of the control printed circuit board **180**. In this way, the through holes **230** and the retaining protrusions **240** are respectively coupled to each other at a plurality of positions (corners, points or locations), and thus the control printed circuit board **180** is stably held in the holder frame **210**.

(98) With this structure, referring to FIGS. **5A** and **5B**, the retaining protrusion **240** is inserted in the through hole **230** so as to securely hold the control printed circuit board **180** during transportation or shorten a manufacturing process by excluding a process of holding the control printed circuit board **180**. As the process is optimized, production energy is less consumed.

(99) The first protrusion **250** of the retaining protrusion **240** may be made of iron, steel or the like metal material and coupled to the bottom of the control printed circuit board **180**. The second protrusion **260** is connected to the bottom of the first protrusion **250**. According to an embodiment of the disclosure, the second protrusion **260** may include a nonconductive flexible material such as rubber, silicon, and urethane, so that the control printed circuit board **180** can be easily coupled to the holder frame **210** and prevented from shaking. Thus, the second protrusion **260** can be considered a flexible portion of the retaining protrusion **240** in one embodiment.

(100) When the recessed portion **262** has a corresponding size (e.g., width) to, for example, the same size as the retaining hole **233**, the recessed portion **262** is inserted in the retaining hole **233** and then firmly pressed against the inner surface (e.g., edge) of the retaining hole **233** such that the

recessed portion **262** is in contact with the inner surface of the retaining hole **233**. Thus, after the recessed portion **262** is inserted in the retaining hole **233**, the outer surface of the recessed portion **262** made of a flexible material is firmly in contact with the inner surface of the retaining hole **233**. Accordingly, the retaining protrusion **240** is in contact with at least a portion of the through hole **230** in which the retaining protrusion **240** is inserted.

(101) As described above, the holding portion **236** protruding inwards to form a concave surface may be formed between the insertion hole **231** and the retaining hole **233**, and the holding portion **236** may be smaller than the recessed portion **262**. Therefore, the recessed portion **262** is flexible and thus is slightly compressed inwards while passing the holding portion **236**, and returns to its original shape after passing the holding portion **236**, thereby being inserted in the retaining hole **233**. The holding portion **236** may prevent (interfere with) the movement of the recessed portion **262** from the retaining hole **233** to the insertion hole **231**.

(102) Further, the recessed portion **262** is nonconductive to prevent current from flowing between the control printed circuit board **180** and the holder frame **210**.

(103) In addition, the second protrusion **260** is made of a nonconductive flexible material to absorb shock and prevent noise even though vibration occurs and causes the control printed circuit board **180** to move when the recessed portion **262** is positioned inside the retaining hole **233**. Of course, the recessed portion **262** is firmly in contact with the retaining hole **233**, thereby offsetting the movement of the control printed circuit board **180**.

(104) Referring to FIGS. 5A and 5B, when the second protrusion **260** is inserted in the insertion hole **231**, the bottom portion **264** is positioned inside the holder frame **210** and the recessed portion **262** is positioned in the insertion hole **231**.

(105) In addition, referring to FIGS. 6A and 6B, when a user grasps and moves the control printed circuit board **180** in a downward direction, the recessed portion **262** is moved toward the retaining hole **233**. In this case, the bottom portion **264** is positioned below the retaining hole **233** inside the holder frame **210**.

(106) When the recessed portion **262** has the same size as the retaining hole **233**, the recessed portion **262** may be firmly held in the retaining hole **233**. Further, the retaining hole **233** is smaller than the bottom portion **264**, and thus the control printed circuit board **180** is prevented from separating upwards.

(107) FIG. 10 discloses another embodiment of the holder member **200**.

(108) Referring to FIG. 10, an insertion hole **231'** and a retaining hole **233'** of a through hole **230'**, and a bottom portion **264'** and a recessed portion **262'** of a second protrusion **260'** may have a triangle (or tree)-shaped cross-section. The insertion hole **231'** and the retaining hole **233'** of the through hole **230'**, and the bottom portion **264'** and the recessed portion **262'** of the second protrusion **260'** may have various cross-sectional shapes such as a circle, and a triangle, and embodiments of the disclosure are not limited thereto.

(109) Repetitive descriptions to the foregoing embodiments of the disclosure will be omitted.

(110) Further, the insertion hole **231'** may be formed to be larger than the retaining hole **233'**. In this case, the bottom portion **264'** may be formed to be smaller than the insertion hole **231'** and larger than the retaining hole **233'**.

(111) The size of the recessed portion **262'** may correspond to the size of the retaining hole **233'**. For example, the size of the recessed portion **262'** may be the same as or slightly smaller than the size of the retaining hole **233'**.

(112) Like the embodiment of FIG. 9, a user moves the control printed circuit board **180** in a downward direction and inserts the bottom portion **264'** and the recessed portion **262'** in the insertion hole **231'**. Because the bottom portion **264'** having a triangular shape is smaller than the insertion hole **231'** having a triangular shape, the bottom portion **264'** is smoothly inserted in the insertion hole **231'** and positioned inside the holder frame **210**.

(113) Then, when a user moves the control printed circuit board **180** downwards in the Y-axis

direction, the recessed portion **262'** is moved to and inserted in the retaining hole **233'**. As described above, the recessed portion **262'** is made of rubber, urethane, silicon, or the like flexible material, and thus passes through a holding portion **236'** formed between the insertion hole **231'** and the retaining hole **233'**. In other words, opposite edges of the recessed portion **262'** having a triangular shape are folded to pass the holding portion **236'**.

(114) After the recessed portion **262'** passes the holding portion **236'**, the opposite edges of the recessed portion **262'** having a triangular shape are unfolded to its original shape and caught in (e.g., connected to) the holding portion **236'**, thereby preventing the recessed portion **262'** from moving from the retaining hole **233'** back to the insertion hole **231'**. The recessed portion **262'** having the triangular shape is caught in the holding portion **236'** and retained in position inside the retaining hole **233'**.

(115) As described above, the holder member **200** according to this embodiment stably holds the control printed circuit board **180** like that of the previous embodiment even though the through hole **230'** and the second protrusion **260'** have a triangular shape.

(116) Meanwhile, FIG. **11** discloses still another embodiment of the holder member **200**.

(117) Referring to FIG. **11**, the holder member **200** according to this embodiment of the disclosure may further include a tapered portion **235** and a clip **270** in addition to the insertion hole **231** and the retaining hole **233** of the through hole **230**. In the embodiment shown in FIG. **11**, the insertion hole **231**, the retaining hole **233**, the recessed portion **262** and the bottom portion **264** may have circular shapes like those of the embodiment shown in FIG. **9**.

(118) The tapered portion **235** may be disposed between the insertion hole **231** and the retaining hole **233**, and may taper in a direction from the insertion hole **231** toward the retaining hole **233**.

(119) The clip **270** may be disposed between the insertion hole **231** and the retaining hole **233**. The clip **270** may be resiliently bent in an inward direction of the holder frame **210** when the recessed portion **262** moves from the insertion hole **231** to the retaining hole **233**, but resiliently stretched to come into contact with the surface of the recessed portion **262** after the recessed portion **262** is inserted in the retaining hole **233**. For example, the clip **270** may contact-support the surface of the recessed portion **262** and prevent the recessed portion **262** from moving from the retaining hole **233** back to the insertion hole **231**, thereby retaining the position of the recessed portion **262**.

(120) Meanwhile, the tapered portion **235** gradually becomes narrower in a direction from the insertion hole **231** toward the retaining hole **233**. The size of the tapered portion **235** may be smaller than the diameter of the retaining hole **233** in the vicinity of the retaining hole **233**. As described above, the recessed portion **262** is made of rubber, silicon, urethane or the like flexible material, and passes the tapered portion **235** while being compressed inwards.

(121) The recessed portion **262** is then contact-supported by the clip **270** and thus prevented from separating from the retaining hole **233**. Further, the holding portion **236** is formed to be smaller than the recessed portion **262** and also serves to prevent the separation of the recessed portion **262**.

(122) With this structure, the holder member **200** according to an embodiment of the disclosure stably holds the control printed circuit board **180**.

(123) The display apparatus according to the disclosure is as follows.

(124) According to an embodiment of the disclosure, a display apparatus may include a display panel; a heat dissipation plate disposed on a rear of the display panel; a source printed circuit board disposed on a rear of the heat dissipation plate, and connected to the display panel by a flexible circuit board; a control printed circuit board disposed on the rear of the heat dissipation plate, connected to the source printed circuit board by a flat cable, and including a retaining protrusion; and a holder member disposed between the heat dissipation plate and the control printed circuit board, and formed with a through hole through which the retaining protrusion passes.

(125) According to some embodiments of the disclosure, the holder member may include a holder frame disposed on the rear of the heat dissipation plate and including the through hole.

(126) According to some embodiments of the disclosure, the through hole may include: an insertion

hole disposed in the holder frame and allowing the retaining protrusion to be inserted therein; a retaining hole disposed in the holder frame, communicating with the insertion hole, and retaining the retaining protrusion; and a holding portion disposed between the insertion hole and the retaining hole and protruding inwards.

(127) According to some embodiments of the disclosure, the retaining hole may be smaller than the insertion hole.

(128) According to some embodiments of the disclosure, the retaining protrusion may include: a first protrusion connected to a lower portion of the control printed circuit board and protruding in a downward direction; and a second protrusion connected to a lower portion of the first protrusion and inserted in the through hole.

(129) According to some embodiments of the disclosure, the second protrusion may include: a recessed portion recessed inwards in a middle portion of the second protrusion; and a bottom portion formed to be larger than the recessed portion in the bottom of the second protrusion.

(130) According to some embodiments of the disclosure, the recessed portion may be formed to have a size corresponding to the retaining hole, and the recessed portion may be formed to be larger than the holding portion.

(131) According to some embodiments of the disclosure, the bottom portion may be formed to be smaller than the insertion hole and larger than the retaining hole.

(132) According to some embodiments of the disclosure, the bottom portion may be shaped corresponding to the insertion hole, and the recessed portion may be shaped corresponding to the retaining hole.

(133) According to some embodiments of the disclosure, the insertion hole and the retaining hole may have a circular shape, the bottom portion and the recessed portion may have a circular shape, the bottom portion may have a smaller diameter than the insertion hole but a larger than the retaining hole, and the recessed portion may have a diameter corresponding to the retaining hole.

(134) According to some embodiments of the disclosure, the insertion hole and the retaining hole may have a triangular shape, the bottom portion and the recessed portion may have a triangular shape, the bottom portion may be smaller than the insertion hole but larger than the retaining hole, and the recessed portion may have a size corresponding to the retaining hole.

(135) According to some embodiments of the disclosure, the recessed portion having the triangular shape may be caught in the holding portion and retained in position inside the retaining hole.

(136) According to some embodiments of the disclosure, the through hole may be disposed between the insertion hole and the retaining hole, and include a tapered portion gradually decreased in size in a direction from the insertion hole toward the retaining hole.

(137) According to some embodiments of the disclosure, the display apparatus may further include a clip disposed between the insertion hole and the retaining hole, wherein the clip is resiliently bent while the recessed portion is moving from the insertion hole to the retaining hole, and resiliently stretched to come into contact with the recessed portion and retain the position of the recessed portion after the recessed portion is inserted in the retaining hole.

(138) According to some embodiments of the disclosure, a direction where the insertion hole and the retaining hole are arranged may correspond to a direction where the flat cable connects the control printed circuit board and the source printed circuit board.

(139) According to some embodiments of the disclosure, the first protrusion may include a nonflexible material, and the second protrusion may include a nonconductive flexible material.

(140) According to some embodiments of the disclosure, the holder member may include an opening formed penetrating an area of the holder frame corresponding to the control printed circuit board, wherein heat generated in the control printed circuit board is dissipated through the opening.

(141) The foregoing descriptions are merely a specific embodiment of the display apparatus.

(142) Therefore, it is readily apparent to a person having ordinary knowledge in the art that various substitutions and modifications can be made without departing from the scope of the disclosure

defined in the appended claims.

(143) According to an embodiment of the disclosure, the display apparatus may be applicable to a mobile device, a video telephone, a smart watch, a watch phone, a wearable apparatus, a foldable apparatus, a rollable apparatus, a bendable apparatus, a flexible apparatus, a curved apparatus, a sliding apparatus, a variable apparatus, an electronic notebook an E-book, a portable multimedia player (PMP), a personal digital assistant (PDA), an MP3 player, a mobile medical device, a desktop personal computer (PC), a laptop PC, a netbook computer, a workstation, a navigation, an in-vehicle navigation, an in-vehicle display, an in-vehicle device, a theater device, a theater display, a television, a wallpaper device, a signage device, a game console, a laptop computer, a monitor, a camera, a camcorder, home appliances, etc.

(144) Although a few embodiments have been described in more detail with reference to the accompanying drawings, the disclosure is not necessarily limited to such embodiments and may be variously modified without departing from the scope of the disclosure. Therefore, embodiments of the disclosure are intended to not limit but describe the technical spirit of the disclosure, and the scope of the disclosure is not limited by these embodiments. Accordingly, the embodiments described above should be understood to be illustrative and non-limiting in all respects. The scope of the disclosure should be construed by the appended claims, and all technical spirits within the equivalent scope fall in the scope of the disclosure.

(145) TABLE-US-00001 DESCRIPTION OF REFERENCE NUMERALS 100: display apparatus 101: cover member 150: display panel 160: heat dissipation plate 170: source printed circuit board 171: first connector 175: flexible circuit board 180: control printed circuit board 181: second connector 185: flat cable 190: adhesive member 200: holder member 210: holder frame 220: opening 230, 230': through hole 231, 231': insertion hole 233, 233': retaining hole 236, 236': holding portion 235: tapered portion 240, 240': retaining protrusion 250: first protrusion 252: coupling portion 260, 260': second protrusion 262, 262': recessed portion 264, 264': bottom portion 270: clip

Claims

1. A display apparatus comprising: a display panel including a front surface and a rear surface that is opposite the front surface; a heat dissipation plate including a front surface and a rear surface, the front surface of the heat dissipation plate on the rear surface of the display panel; a source printed circuit board on the rear surface of the heat dissipation plate, the source printed circuit board connected to the display panel via a flexible circuit board; a control printed circuit board on the rear surface of the heat dissipation plate and connected to the source printed circuit board via a flat cable, the control printed circuit board comprising a retaining protrusion that protrudes towards the rear surface of the display panel; and a holder member between the heat dissipation plate and the control printed circuit board, the holder member comprising a holder frame on the rear surface of the heat dissipation plate and the holder frame including at least one through hole through which the retaining protrusion is inserted, wherein the at least one through hole comprises: an insertion hole in the holder frame; a retaining hole in the holder frame; and a holding portion between the insertion hole and the retaining hole, the holding portion having a width that is less than a width of the insertion hole and a width of the retaining hole, wherein the retaining protrusion is configured to be inserted through the insertion hole and moved into the retaining hole through the holding portion.

2. The display apparatus of claim 1, wherein the at least one through hole includes a plurality of through holes at edges of the holder frame.

3. The display apparatus of claim 1, wherein the retaining hole is smaller than the insertion hole.

4. The display apparatus of claim 3, wherein the retaining protrusion comprises: a first protrusion connected to a lower portion of the control printed circuit board, the first protrusion protruding in a

downward direction away from the lower portion of the control printed circuit board; and a second protrusion connected to a lower portion of the first protrusion, the second protrusion inserted in the retaining hole of the at least one through hole.

5. The display apparatus of claim 4, wherein the second protrusion comprises: a bottom portion at a bottom end of the second protrusion; and a recessed portion between the bottom portion of the second protrusion and the first protrusion, the recessed portion having a width that is less than a width of the bottom portion and a width of the first protrusion.

6. The display apparatus of claim 5, wherein the recessed portion has a size corresponding to a size of the retaining hole, and the size of the recessed portion is larger than a size of the holding portion.

7. The display apparatus of claim 6, wherein the recessed portion is flexible and is compressed while disposed in the holding portion such that the recessed portion has a size that is less than a size of the recessed portion while the recessed portion is inserted in the retaining hole.

8. The display apparatus of claim 6, wherein a size of the bottom portion is smaller than a size of the insertion hole and is larger than the size of the retaining hole.

9. The display apparatus of claim 8, wherein the bottom portion has a shape that corresponds to a shape of the insertion hole, and the recessed portion has a shape that corresponds to a shape of the retaining hole.

10. The display apparatus of claim 9, wherein the insertion hole, the retaining hole, the bottom portion, and the recessed portion have a circular shape, wherein the bottom portion has a diameter that is smaller than a diameter of the insertion hole but larger than a diameter of the retaining hole, and the recessed portion has a diameter corresponding to the diameter of the retaining hole.

11. The display apparatus of claim 5, wherein the insertion hole, the retaining hole, the bottom portion, and the recessed portion have a triangular shape, wherein a size of the bottom portion is smaller than a size of the insertion hole but larger than a size of the retaining hole, and a size of the recessed portion corresponds to the size of the retaining hole.

12. The display apparatus of claim 11, wherein the recessed portion having the triangular shape is connected to the holding portion and retained inside the retaining hole.

13. The display apparatus of claim 5, wherein the at least one through hole comprises a tapered portion that tapers in a direction from the insertion hole toward the retaining hole.

14. The display apparatus of claim 13, further comprising: a clip disposed between the insertion hole and the retaining hole, wherein the clip is configured to be bent while the recessed portion moves from the insertion hole to the retaining hole, and contacts the recessed portion to retain a position of the recessed portion while the recessed portion is in the retaining hole.

15. The display apparatus of claim 1, wherein a direction in which the insertion hole and the retaining hole extend corresponds to a direction in which the flat cable extends to connect the control printed circuit board and the source printed circuit board.

16. The display apparatus of claim 4, wherein the first protrusion comprises a non-flexible material, and the second protrusion comprises a non-conductive flexible material.

17. The display apparatus of claim 1, wherein the holder member comprises an opening through an entire thickness of an area of the holder frame corresponding to the control printed circuit board.

18. A display apparatus comprising: a display panel including a front surface and a rear surface that is opposite the front surface; a heat dissipation plate including a front surface and a rear surface and the front surface of the heat dissipation plate is on the rear surface of the display panel, the heat dissipation plate configured to dissipate heat from the display panel; a holder frame on the heat dissipation plate, the holder frame including a plurality of through holes; and a control printed circuit board on the holder frame, the control printed circuit board including a plurality of retaining protrusions that are each inserted into a corresponding through hole from the plurality of through holes and in contact with at least a portion of the corresponding through hole such that the control printed circuit board is connected to the holder frame; wherein each of the plurality of through holes comprises: an insertion hole in the holder frame; a retaining hole in the holder frame having a

width that is wider than a width of the insertion hole; and a holding portion between the insertion hole and the retaining hole, the holding portion having a width that is less than the width of the insertion hole and the width of the retaining hole, wherein each retaining protrusion is configured to be inserted through the insertion hole of the corresponding through hole and moved into the retaining hole of the corresponding through hole through the holding portion of the corresponding through hole.

19. The display apparatus of claim 18, wherein each of the plurality of retaining protrusions comprises: a flexible portion having a first end, a second end, and a recessed portion between the first end and the second end of the flexible portion, the recessed portion having a width that is less than a width of the first end and a width of the second end.

20. The display apparatus of claim 19, wherein the width of the recessed portion is a same as the width of the retaining hole such that the recessed portion is in contact with an edge of the retaining hole while the retaining protrusion is in the retaining hole, and the width of the second end of the flexible portion is wider than the width of the retaining hole.

21. The display apparatus of claim 19, wherein the width of the recessed portion is wider than the width of the holding portion, wherein the recessed portion is compressed while in the holding portion such that the width of the recessed portion is less than the width of the recessed portion while the recessed portion is inserted in the retaining hole.

22. The display apparatus of claim 18, wherein the holder frame includes an opening through an entire thickness of the holder frame and the control printed circuit board overlaps the opening.
